MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

Texts

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*
- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*
- **Conrad** — Conrad, *The Gaussian Integers*

Schedule

Date		Topic	Reading	Due
Thu 20 Aug		Course launch; divisibility and the division algorithm	KvP §1.1-1.2	
Tue 25 Aug		Primes, the Fundamental Theorem of Arithmetic, gcd and lcm	KvP §1.3-1.5	
Thu 27 Aug		Semigroups, monoids, groups — all of them Z/nZ	KvP §2.1	
Tue 1 Sep		Subgroups; the unit group (Z/nZ)*	KvP §2.2	Problem Set 1
Thu 3 Sep	**Q&A**	Defense 1		
Tue 8 Sep		Group homomorphisms; the order of an element	KvP §2.3	
Thu 10 Sep		Cyclic groups; primitive roots	KvP §2.3	
Tue 15 Sep		Cosets and Lagrange's theorem	KvP §2.4	Problem Set 2
Thu 17 Sep	**Q&A**	Defense 2		
Tue 22 Sep		Normal subgroups and quotient groups	KvP §2.5	
Thu 24 Sep		The homomorphism theorem	KvP §2.6	
Tue 29 Sep		Euler's theorem, fast modular exponentiation, and RSA end to end	KvP App. B	
Thu 1 Oct	**Exam**	Midterm Exam 1 — Segment 1		
Tue 6 Oct		Rings and subrings	KvP §3.2	
Thu 8 Oct		Ring homomorphisms	KvP §3.3	
Thu 15 Oct		Ideals	KvP §3.3	
Tue 20 Oct		Quotient rings	KvP §3.3	Problem Set 3
Thu 22 Oct	**Q&A**	Defense 3		
Tue 27 Oct		Fields and skew fields; F_p	KvP §3.4	
Thu 29 Oct		Integral domains and the field of fractions	KvP §3.5	
Tue 3 Nov		Unique factorization: UFDs	KvP §3.7	Problem Set 4
Thu 5 Nov	**Q&A**	Defense 4		
Tue 10 Nov		PIDs and Euclidean domains	KvP §3.7	

Date		Topic	Reading	Due
Thu 12 Nov		Complex numbers: arithmetic, the plane, polar form, roots of unity	Beardon §3.1-3.2, §3.5	
Tue 17 Nov		Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points	Beardon Ch. 13	Problem Set 5
Thu 19 Nov	**Q&A**	Defense 5		
Tue 24 Nov		The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey	Conrad	
Tue 1 Dec	*Review*	Review — the exam pool		
Thu 3 Dec	**Exam**	Midterm Exam 2 — Segment 2 weighted		